

=== GNB (1780953769) ===

```
cu_cp:
  amf:
    addr: 10.53.1.2
    port: 38412
    bind_addr: 10.53.1.1
    supported_tracking_areas:
      - tac: 7
    plmn_list:
      - plmn: "00101"
    tai_slice_support_list:
      - sst: 1 # Extracted from 3GPP TS 23.501, Table 5.15.2.2-1: Standardized S-NSSAI values for eMBB.
    inactivity_timer: 2700
ru_sdr:
  device_driver: zmq
  device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=11.52e6
  srate: 11.52
  tx_gain: 75
  rx_gain: 75
cell_cfg:
  dl_arfcn: 368500
  band: 3
  channel_bandwidth_MHz: 10
  common_scs: "15kHz"
  ssb_scs: "15kHz"
  plmn: "00101"
  tac: 7
  pdccch:
    common:
      ss0_index: 0
      coreset0_index: 6
    dedicated:
      ss2_type: common
      dci_format_0_1_and_1_1: false
  prach:
    prach_config_index: 1
  pdsch:
    mcs_table: qam64
  pusch:
    mcs_table: qam64
log:
  filename: /tmp/gnb.log
  all_level: info
  hex_max_size: 0
pcap:
  mac_enable: false
  mac_filename: /tmp/gnb_mac.pcap
  ngap_enable: true
  ngap_filename: /tmp/gnb_ngap.pcap
  e2ap_enable: true
  e2ap_du_filename: /tmp/gnb_du_e2ap.pcap
  e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap
  e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap
metrics:
  layers:
    enable_rlc: true
    enable_sched: true
  periodicity:
    du_report_period: 400
```

=== UE (1780953769) ===

```
[rf]
freq_offset = 0
tx_gain = 50
rx_gain = 40
srate = 11.52e6
nof_antennas = 1

device_name = zmq
device_args = tx_port=tcp://127.0.0.1:2001,rx_port=tcp://127.0.0.1:2000,base_srate=11.52e6

[rat.eutra]
dl_eurfcn = 2850
nof_carriers = 0

[rat.nr]
bands = 3
nof_carriers = 1

max_nof_prb = 52
nof_prb = 52

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 112233445566778899aabbccddeeff00
k = 00aabbccddeeff112233445566778899
imsi = 0010100000000001
imei = 353490069873319

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
netns = ue1
ip_devname = tun_srsue
ip_netmask = 255.255.255.0

[gui]
enable = false
```

=== DOCKER (1780953769) ===

```
#
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#
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#
```

services:

5gc:

```
  container_name: open5gs_5gc
  build:
    context: open5gs
    target: open5gs
    args:
      OS_VERSION: "22.04"
      OPEN5GS_VERSION: "v2.7.0"
  environment:
    - MONGODB_IP=127.0.0.1
    - OPEN5GS_IP=10.53.1.2
    - UE_IP_BASE=10.45.0
    - UPF_ADVERTISE_IP=10.53.1.2
    - DEBUG=false
```

SUBSCRIBER_DB=001010000000001,00aabbccddeeff112233445566778899,opc,112233445566778899aabbccddeeff00,8000,9,10.45.0.2

```
- NETWORK_NAME_FULL=srsRAN
- NETWORK_NAME_SHORT=srsRAN
- TZ=Europe/Madrid
```

privileged: true

ports:

```
- "9898:9999/tcp"
```

Uncomment port to use the 5gc from outside the docker network

```
# - "38412:38412/sctp"
```

```
# - "2152:2152/udp"
```

command: 5gc -c open5gs-5gc.yml

healthcheck:

```
  test: [ "CMD-SHELL", "nc -z 127.0.0.20 7777" ]
```

interval: 3s

timeout: 1s

retries: 60

networks:

ran:

```
  ipv4_address: ${OPEN5GS_IP:-10.53.1.2}
```

gnb:

container_name: srsran_gnb

Build info

image: srsran/gnb

build:

context: ..

dockerfile: docker/Dockerfile

```

args:
  OS_VERSION: "24.04"
# privileged mode is required only for accessing usb devices
privileged: true
# Extra capabilities always required
cap_add:
  - SYS_NICE
  - CAP_SYS_PTRACE
volumes:
  # Access USB to use some SDRs
  - /dev/bus/usb/:/dev/bus/usb/
  # Sharing images between the host and the pod.
  # It's also possible to download the images inside the pod
  - /usr/share/uhd/images:/usr/share/uhd/images
  # Save logs and more into gnb-storage
  - gnb-storage:/tmp
# It creates a file/folder into /config_name inside the container
# Its content would be the value of the file used to create the config
configs:
  - gnb_config.yml
  - gnb_compose_config.yml
# Customize your desired network mode.
# current network configuration creates a private network with both containers attached
# An alternative would be `network: host`. That would expose your host network into the container. It's
the easiest to use if the 5gc is not in your PC
networks:
  ran:
    ipv4_address: ${GNB_IP:-10.53.1.1}
  metrics:
    ipv4_address: 172.19.1.3
# Start GNB container after 5gc is up and running
depends_on:
  5gc:
    condition: service_healthy
# Command to run into the final container
command: gnb -c /gnb_config.yml -c /gnb_compose_config.yml

configs:
gnb_config.yml:
  file: gnb_config.yml # Path to your desired config file
gnb_compose_config.yml:
  content: |
    cu_cp:
      amf:
        addr: ${OPEN5GS_IP:-10.53.1.2}
        bind_addr: 0.0.0.0
    metrics:
      autostart_stdout_metrics: true
      enable_json: true
    remote_control:
      bind_addr: 0.0.0.0
      enabled: true

volumes:
  gnb-storage:

networks:
  ran:
    ipam:
      driver: default
      config:
        - subnet: 10.53.1.0/24
  metrics:
    ipam:
      driver: default
      config:
        - subnet: 172.19.1.0/24

```

=== NOTAS (1780953769) ===

- PLMN (MCC 001, MNC 01, TAC 7) y IMSI (001010000000001) consistentes en gNB, UE y Docker.
- dl_arfcn 368500 y Banda 3 son coherentes (REGLA 2).
- Indentación de 2 espacios y formato "15kHz" para common_scs y ssb_scs aplicados (REGLA 3).
- ssb_scs se añadió después de common_scs con el mismo valor y nivel de indentación (REGLA 3c).
- Puertos ZMQ 2000 y 2001 cruzados entre gNB y UE (REGLA 4).
- bind_addr configurado según REGLA 5 (10.53.1.1 en gNB config, 0.0.0.0 en docker-compose override).
- Bloque ru_sdr copiado inmutablemente del CAG (REGLA 6a).
- srates 11.52e6 es compatible con ancho de banda de 10 MHz (REGLA 6b) y consistente entre gNB y UE (REGLA 6c).
- IMSI, K y OPC en SUBSCRIBER_DB y UE son idénticos y en el orden correcto (REGLA 7).
- IP estática 10.45.0.2 en SUBSCRIBER_DB es consistente con UE_IP_BASE=10.45.0.
- SST 1 para eMBB añadido con comentario de referencia.
- inactivity_timer configurado a 2700 segundos (45 minutos).
- Métricas RLC y scheduler DU habilitadas con periodo de 400 ms.
- Capturas PCAP de NGAP habilitadas.
- max_nof_prb y nof_prb del UE ajustados a 52 para 10 MHz BW y 15 kHz SCS.